INCH-POUND

MIL-R-23417/19A
25 April 1991

SUPERSEDING
MIL-R-23417/19
22 September 1967

MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL, AC
TYPE 23R58N4A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Electrical Resolvers described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-R-23417.

TABLE I. Requirements.

Requirements	Value	Unit	Tolerance
Excitation winding	Stator		
Rated frequency	400	Hz	±1%
Input voltage:			
Maximum	100	volts (rms)	±1%
Test	50	volts (rms)	±1%
Minimum	1.0	volts (rms)	±1%
Primary current	8.5	milliamps	nominal
Primary power	0.063	watts	nominal
D.C. resistance:			
Rotor	230	ohms	±15%
Stator	130	ohms	±15%
Impedance:			
Zro	950+j5900	ohms	R±15%+jx±10%
Zso	870+j5790	ohms	R±15%+jx±10%
Zrs	360+j528	ohms	nominal
Transformation ratio:			
Output/input	0.975	----	±1%
T.R. variation			
due to voltage	0.05	percent	maximum

AMSC - N/A FSC 5990

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:43Z
Check the source to verify that this is the current version before use.

TABLE I. Requirements - Continued.

Requirements	Value	Unit	Tolerance
Equality of transformation ratios: Output/input	0.08	percent spread	maximum
Phase shift: Output/input	1.2	degrees	±0.5
Phase shift variation: due to voltage	±0.5	degrees	maximum
Interaxis error	±3.0	minutes	maximum
Functional error	±0.05	percent	maximum
Null voltage: Fundamental	20	mV(rms) output	maximum
Total	30	mV(rms) output	maximum
Shift of resolver zero: With voltage	±1.0	minutes	maximum
With frequency	±0.5	minutes	maximum
Harmonic distortion	0.1	percent	maximum
Temperature rise	20	degrees C	maximum
Mechanical breakway torque:	0.2	oz-in	maximum
@ -55°C	0.4	oz-in	maximum
Radial play	0.0010	inches	maximum
End play	0.0030	inches	±0.002
Resonant frequency	30	kHz	minimum
Low corner frequency	8.3	Hz	maximum

NOTE:

1. A voltage value of 1 to 100 volts shall be entered in the "volts" designation space on the identification plate.

TABLE II. Part identifying number variant characteristics.

Part Identifying No. 1/	A ±.010	B ±.010	G Maximum	Shaft Style	Terminal End	U ±.0000 -.0005
M23417/19-01A	.676	.724	3.600	Spline	Terminal Board	.2405

1/ Part identifying number M23417/19-01A shall be the NATO standard.

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

Source: https://assist.dla.mil -- Downloaded: 202610-19T16:43Z
Check the source to verify that this is the current version before use.

TABLE III. Part identifying number cross-reference. 1/

SUPERSEDED PART NUMBERS	NEW PART IDENTIFYING NUMBERS
MIL-R-23417/19	
M23417/19-01	M23417/19-01A

1/ All line items shown in Table III refer to equivalent and interchangeable electrical resolvers of the same type designation modification. Part number changes do not affect form, fit or function of the electrical resolvers listed therein.

Custodians:
Army - AR
Navy - AS
Air Force - 85

Review activities:
Navy - OS, EC
Air Force - 99

User activities:
Army - MI, AV, ME, AT
Navy - CG, MC
Air Force - 19, 14

Preparing activity:
Army - AR

(Project 5990-0386)

Source: https://assist.dla.mil -- Downloaded: 202640-19T16:43Z
Check the source to verify that this is the current version before use.